

Solar Panel Soiling Detection Web App

Important setup instructions for running the project on another laptop

1. Before starting

Extract the ZIP file first. Do not run the project directly from inside the ZIP file. Use a simple folder path:

```
D:/SLT/SOLA/solar-soiling-web-app
```

Open the extracted **solar-soiling-web-app** folder using VS Code.

2. Required software

Software	Purpose
Python 3.10 or newer	Run Flask backend and ML model
VS Code	Open and run the project
Live Server extension	Run the HTML/CSS/JS frontend

3. Project structure

Confirm this folder structure after extracting:

```
solar-soiling-web-app
|-- backend
|   |-- app.py
|   |-- requirements.txt
|   |-- model
|   |-- uploads
|   |-- results
|   |-- yolov5
|-- frontend
|   |-- index.html
|   |-- style.css
|   |-- script.js
```

Important: the trained model file must be inside backend/model. The model path in app.py must match the real model file name.

4. Backend setup

Open a VS Code terminal and run these commands one by one:

```
cd "D:/SLT/SOLA/solar-soiling-web-app/backend"
python -m venv .venv
.venv/Scripts/activate
pip install -r requirements.txt
python app.py
```

Backend should run on:

```
http://127.0.0.1:5000
```

Expected browser test message:

```
{
  "message": "Solar Panel Soiling Detection Backend is running"
}
```

5. Frontend setup

This frontend uses normal HTML, CSS, and JavaScript. Do not run npm install. There is no need for Node.js setup.

In VS Code, open **frontend/index.html**. Right-click inside the file and select **Open with Live Server**.

Frontend usually opens on:

`http://127.0.0.1:5500`

6. Correct running order

The backend and frontend must run at the same time.

Step	Action
1	Start Flask backend using python app.py
2	Open frontend using Live Server
3	Upload a solar panel image from the web page
4	Click the detection button
5	Backend processes the image and returns result

7. Common problems

Error	Fix
requirements.txt not found	You are not inside backend folder. Run: <code>cd backend</code>
127.0.0.1:5000 refused to connect	Backend is not running. Start it using <code>python app.py</code> .
127.0.0.1:5500 refused to connect	Live Server is not running. Open <code>frontend/index.html</code> with Live Server.
POST /predict 500 error	Backend crashed during prediction. Check model path, packages, or Python traceback.
favicon.ico 404	Not serious. Browser is only asking for a tab icon.

8. Important notes

- Keep the terminal running after starting **python app.py**. If the terminal is closed, backend stops.
- Use Live Server for the frontend. Do not use npm commands for this frontend.
- If the project path has spaces, always use quotation marks in PowerShell commands:
`cd "D:/SLT/SOLA/solar-soiling-web-app/backend"`
- This is a development/demo version. It is suitable for testing and supervisor demonstration.

Prepared for project handover/demo setup.